Epigenetics and Stress Reactivity

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No AI tools were used in any way to help me develop the ideas for or writing of this paper.

Part One: Summary of Concept

Histone modification is the process in which a protein causes a specific chemical change in the histone, which is a core of proteins that is wrapped tightly in a DNA chain. When the histone experiences a chemical change, it forces the DNA coil to open, which ultimately changes the activity of the associated genes.

In order to study histone modification, Allis used the *Tetrahymena* microbe, which has a set of genes that are "turned on" and a set of genes that are "turned off." He realized that there was a different histone associated with each set of genes. Allis found that altering histones caused the DNA coil to open and consequently changed gene activity without affecting the DNA sequence.

Reinberg, Berger, and Liebig studied epigenetics in ant colonies. Reisenberg and Berger were two epigeneticists who collaborated with Liebig, an entomologist studying ant behavior. The body structure, behavior, and social role that ants take in a colony is usually determined by signals from the social and physical environment. Although two ants may have the same genome, they may have completely different genes turned on and off based on environmental factors.

By inducing histone modification in a developing ant, researchers were able to cause a change in the ant's identity within the caste system, for example, from queen to major. This change in identity is marked by physiological and behavioral changes.

Additionally, histone modification can be recorded in a lineage of cells' "memory". Essentially, histone modifications can be passed from a parent cell to its daughter cells during cell division. Allis identified many histone-modifying proteins that had the power to modify histones, and consequently, gene activity. He found that the various proteins interact with one another and overlay information on the genome. They enable the cell to create a team of other cells out of the same genes and pass on “memories” to their descendants. These epigenetic codes can either make a cell more active or make a cell inactive.

Hongerwinter was an acute famine in the Netherlands during World War II that left many surviving children with chronic health conditions. Moreover, the children and grandchildren of men and women exposed to the famine were found to have higher rates of illness than the general population. While it’s impossible for genes in an entire population to change in the span of two generations, this provides evidence for the theory that metabolic stress, or at least the memory of it, is able to be inherited.

Part Two: Connect to Literature

Roth et al., (2009) studied the epigenetic influence of early maltreatment on methylation of Brain Derived Neurotrophic Factor (BDNF) DNA in rats. BDNF is a protein that plays a key role in neuronal survival and growth and has important implications for depression and stress.

This study manipulated early-life adversity by exposing some infant rats to maltreatment from stressed caretakers during their first postnatal week (maltreatment condition). Other pups were exposed to a non-biological caretaker that did not display stressed/abusive behavior (cross-fostering condition). The control group was composed of pups raised by their non-stressed biological mom. The researchers regularly measured BDNF DNA methylation and gene expression in the prefrontal cortex and hippocampus throughout the subjects’ life span, as well as in the subject’s offspring.

The researchers found that rats who were exposed to maltreatment displayed decreased levels of BDNF gene expression in the prefrontal cortex, but no significant change in BDNF gene expression in the hippocampus. They also found altered BDNF DNA methylation in the following generation of infants descending from the maltreatment group.

This study illustrates the epigenetic influence of early maltreatment on methylation of BDNF and gene expression. These changes in DNA were inherited by the offspring of the maltreatment group, supporting the theory that epigenetic modification can be passed down generations. This data supports the Hongerwinter phenomenon outlined in Mukherjee's (2016) article on epigenetics. Mukherjee describes how children exposed to the famine demonstrated increased chronic health issues. This is a similar example of early-life adversity causing physiological changes in the brain and body, and being passed onto offspring.

Part Three: Application

Adult attachment styles refer to the different ways in which humans react to the maintenance and loss of affective bonds (Erkoreka et al., 2021). Attachment styles are influenced by both genetic and environmental factors, and in turn influence cognition and emotion in individuals (Ein-Dor et al., 2018). Research on the epigenetic mechanisms of attachment styles is limited. However, Ein-Dor et al., (2018) found evidence that epigenetic modification of OXTR (oxytocin receptor) and NR3C1 (glucocorticoid receptor) gene promoter was associated with attachment avoidance, a type of insecure attachment style. Oxytocin has important implications for the reliance on social behavior during stress, while glucocorticoid has important implications for stress regulation via the HPA axis. Individuals with an attachment avoidance have lower OXTR and higher NR3C1 gene promoter methylation relative to individuals with secure attachment (Ein Dor et al., 2018). This provides evidence for the epigenetic basis of attachment styles in humans. It is clear that more research and funding is needed to establish a relationship between epigenetics and attachment styles. It would be beneficial to study the outcome of manipulating other target genes in addition to OXTR and NR3C1. There is also a lack of research on attachment styles in animals, such as rodents. Using animal models, it may be possible to study the heritability of attachment styles.

References

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